68606 Solid-State Science and Quantum Devices

Laboratory Manual
Spring 2025

Name:

Student ID:

CONTENTS

INTRODUCTION

Laboratory work is a vital component of Solid-State Science and Quantum Devices, providing hands-on experience with experimental methods in solid state physics and devices. The labs are designed to develop your skills in conducting reliable measurements, analysing and interpreting data and applying critical thinking. While some experiments closely relate to lecture content, others may explore broader concepts.

Each experiment is intended to be completed within a 2.5-hour session. You are required to record your work as you progress in a **laboratory portfolio**, which forms part of your assessment.

What to bring:

Please bring a notebook and a laptop to record measurements, data and observations. After each session, please complete your data analysis and summarise your findings in your portfolio.

SAFETY IN LABORATORIES

Safety in the lab is a shared responsibility. Students must take care to avoid injury to themselves and others, and to prevent damage to equipment. The following safety rules must be observed:

- No eating, drinking or smoking in the laboratories.
- Wear **covered shoes, a lab coat and safety goggles.**
- Some experiments involve **lasers. Never look directly into the laser beam, as severe eye damage can result.**
- When handling **liquid nitrogen, wear cryogenic gloves and safety glasses** to avoid serious burns.
- In the event of an accident, report to a staff member or security officer. For emergencies, contact UTS Security by dialling **6** on campus phones or **1800 279 559** (24 hours).

LOGISTICS

1. You will work in **pairs**, but **each student must submit their own portfolio** individually for assessment.
2. What to bring: a notebook and/or laptop to record your experimental setup, measurements and observations during each lab session.
3. Each lab group will complete a different experiment each week. Please refer to the group allocations below for your weekly experiment schedule.
4. **Group A1 to A9** will attend the morning session from **9:00 am – 11:30 am**, and **Group B1 to B9** will attend the afternoon session from **12:30 pm – 3:00 pm**.

Group	**Week 3** **11 Aug**	**Week 4** **18 Aug**	**Week 5** **25 Aug**
A1	Exp 1	Exp 2	Exp 3
A2	Exp 1	Exp 2	Exp 3
A3	Exp 1	Exp 2	Exp 3
A4	Exp 2	Exp 3	Exp 1
A5	Exp 2	Exp 3	Exp 1
A6	Exp 2	Exp 3	Exp 1
A7	Exp 3	Exp 1	Exp 2
A8	Exp 3	Exp 1	Exp 2
A9	Exp 3	Exp 1	Exp 2

Group	**Week 3**	**Week 4**	**Week 5**
B1	Exp 1	Exp 2	Exp 3
B2	Exp 1	Exp 2	Exp 3
B3	Exp 1	Exp 2	Exp 3
B4	Exp 2	Exp 3	Exp 1
B5	Exp 2	Exp 3	Exp 1
B6	Exp 2	Exp 3	Exp 1
B7	Exp 3	Exp 1	Exp 2
B8	Exp 3	Exp 1	Exp 2
B9	Exp 3	Exp 1	Exp 2

Staff

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Lab manager: Anoj Poudyal (Email: Anoj.Poudyal@uts.edu.au)

Laboratory Portfolio

The laboratory portfolio is a record of your experimental work and an essential part of learning how to conduct and communicate science effectively. It should include your write-up of three experiments, documented clearly in a single portfolio. While the format does not need to follow that of a formal report, **each entry should contain enough detail for someone else to understand your methods, interpret your results, and evaluate your conclusions.**

Entries should be clear, logical and well-organised. Think of your portfolio as a guide that another student—or your future self—could use to repeat the experiments. **Table 1** outlines the key elements expected for each experiment.

Table 1. Suggested Structure for Each Experiment

Section	Description
Date	Record the date of the experiment for reference and tracking.
Title	Clearly identify the topic or objective of the experiment.
Aim	Briefly state what the experiment is trying to achieve.
Apparatus/Setup	Provide a simple, clearly labelled diagram of the experimental setup.
Measurements	Present data in well-organised tables with appropriate units and headings.
Graphs	Include relevant graphs with labelled axes and titles. Graphs from OriginLab may be inserted as needed.
Data analysis	Show your calculations and formulas used. Account for uncertainties where applicable.
Discussion and conclusions	Relate your findings to the experimental aim, discuss results and note any limitations or sources of error.

Assessment Criteria (Total: 30 Marks)
Each of the three experiments is worth 10 marks, based on the following:

- **Organisation and Presentation (2 marks)**
  – Is the portfolio well-structured and easy to follow?
- **Measurement and Data Analysis (6 marks)**
  – Are the data and results clearly presented?
  – Is the analysis correct, including calculations and graphs?
  – Are the results interpreted appropriately?
- **Discussion (2 marks)**
  – Are discussion and conclusion logical and supported by evidence?

Note: For Experiment 1, Parts A and B will each contribute 5 marks.

Experiment 1A: Quantum Sensing with Spins in Diamond

Aims

- Describe the optical and spin properties of the nitrogen-vacancy (NV) centre in diamond
- Utilise an NV magnetometer to measure magnetic fields

Equipment List

- NV magnetometer
- Quantum sensing controller with microwave generator
- Advanced Quantum software (for data acquisition and control of the sensing controller)
- Ruler

Theoretical Background

Nitrogen-Vacancy Centre

The nitrogen vacancy (NV) centre is a point defect in the crystal lattice of diamond, where two carbon atoms are replaced by a nitrogen atom and a vacancy (an empty lattice site). This point defect imparts unique electronic and optical properties that are exploited for various quantum measurements, including quantum magnetometry via the Zeeman effect.

The NV centre contains two unpaired electron, each with a spin quantum number (s) of ½. These electrons combined to form a total spin $S = 1$, resulting in a triplet state (Figure 1). This triplet state consists of three distinct spin levels: $m_s = 0$, $m_s = +1$ and $m_s = -1$.

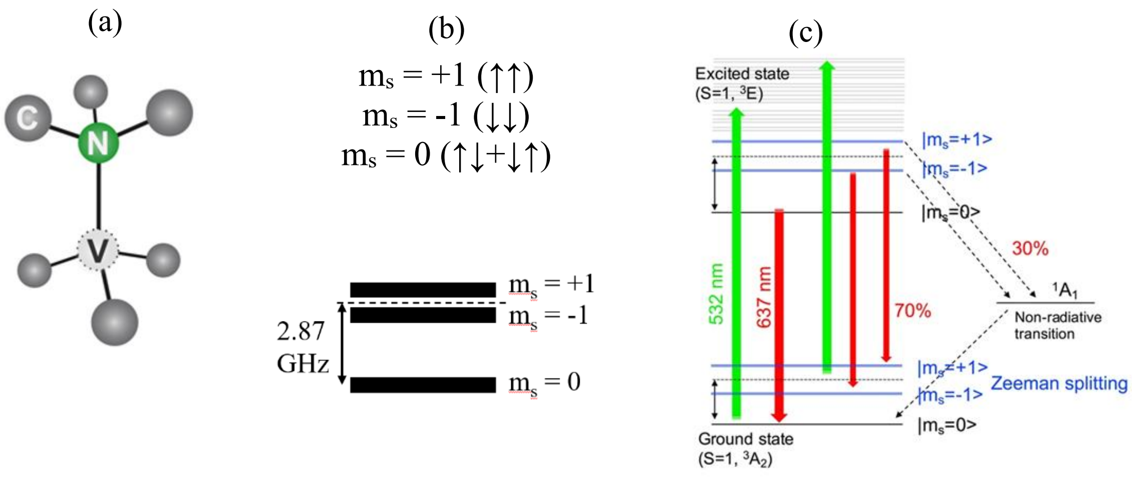


Figure 1: (a) Schematic representation of the NV centre in diamond. (b) Spin triplet states of the NV centre, separated by 2.87 GHz. (c) Energy level diagram showing excitation by a green laser and subsequent red fluorescence emission. Transitions from the ground state $m_s = 0$ to the $m_s = \pm 1$ states can be driven by microwave radiation.

The spin levels $m_s = +1$ and $m_s = -1$ are split from the $m_s = 0$ state by 2.87 GHz due to spin-spin interactions. When the NV centre is irradiated with a green laser, the electron is excited to an excited state and subsequently decays, emitting a photon. The electron predominantly decays to the $m_s = 0$ state. By applying a microwave frequency resonant with the energy difference (2.87 GHz), we can drive transitions between the $m_s = 0$ and $m_s = \pm 1$ levels. Electrons excited to $m_s = \pm 1$ excited states have a higher probability of non-radiative decay (no photon emission). By sweeping through a range of microwave frequencies, we can detect reductions in photon emission corresponding to transitions between the $m_s = 0$ and $m_s = \pm 1$ states.

Zeeman Effect

The Zeeman Effect describes how the energy levels of electrons are split in the presence of a magnetic field. This interaction is described by the following equation:

$$\Delta E = g\mu_B B m_s \tag{1}$$

Where ΔE is the change in energy of the electron due to the magnetic field, g is the dimensionless Landé g-factor (approximately 2 for an electron), μ_B is the Bohr magneton (9.274×10^{-24} J/T), B is the strength of the magnetic field and m_s is the spin quantum number.
To convert this energy change to into frequency space, we use the relation:

$$\Delta E = h\Delta\nu \tag{2}$$

Where h is Planck's constant (6.626×10^{-34} J/Hz) and $\Delta\nu$ is the change in frequency due to a magnetic field. Substituting for ΔE in the frequency space, we get:

$$\Delta\nu = \frac{g\mu_B}{h} B m_s \tag{3}$$

The frequency shift can be simplified to

$$\Delta\nu = \gamma_e B m_s \tag{4}$$

Where $\gamma_e = \frac{g\mu_B}{h} = 28.024$ GHz/T is called gyromagnetic ratio. For the spin state $m_s = 0$, the frequency shift $\Delta\nu$ is zero and hence independent of the magnetic field. For the spin states $m_s = +1$ and $m_s = -1$, the frequency shift depends linearly on the applied magnetic field, leading to Zeeman splitting (Figure 2).

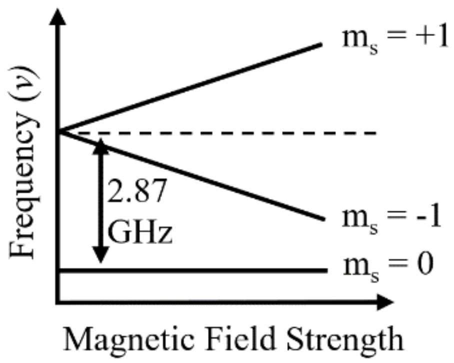


Figure 2: Transitions to $m_s = \pm 1$ spin states can be induced using microwave radiation at approximately 2.87 GHz. When a magnetic field is applied along the NV axis, it lifts the degeneracy of the $m_s = \pm 1$ states, resulting in Zeeman splitting.

Since we measure the absolute value of the frequency shift for both $m_s = +1$ and $m_s = -1$, we can simplify (4) to:

$$\Delta\nu = \gamma_e B \tag{5}$$

Experimental Work

Part 1. Zeeman Splitting Measurement

a) Initial setup and measurement

- To begin measuring Optically Detected Magnetic Resonance (ODMR), click “Enable Laser” then select “ODMR”.
- Set the laser power to approximately 60% and microwave power between 5 and 20 dBm.
- In the “Frequency” setting, select “Center/Span” and set:
    - Centre Frequency to 2.87 GHz (resonant frequency of the NV centre)
    - Frequency Span to around 60 MHz (this will perform a frequency sweep of 60 MHz centred around the resonant frequency of the NV centre)
    - Frequency Delta to 0.5 MHz
- Click “Run Measurement”.
- Allow the system to stabilise
- Continue sweeping until the ODMR spectrum clearly displays two distinct dips, corresponding to the spin transitions: $m_s = 0 \rightarrow -1$ and $m_s = 0 \rightarrow +1$. Figure 3 shows an example of the appropriate settings and the observed ODMR spectrum with Zeeman splitting.

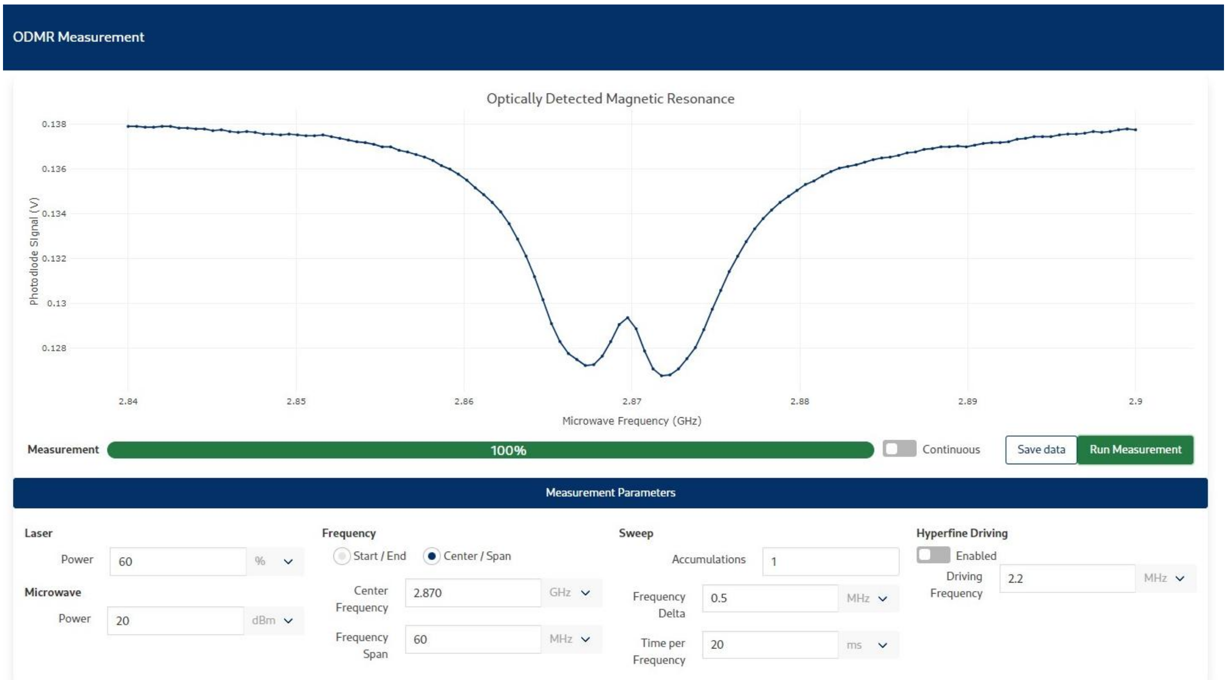


Figure 3: ODMR control panel showing example settings for the measurement.

b) Collecting ODMR spectra

- Two slightly different magnet setups (A and B) are available as shown in Figure 4.
- If you are using setup A:
  - o Position the magnet approximately 5 cm from the tip of the NV magnetometer head as shown in Figure 4A.
  - o Ensure that the magnet is aligned along the central axis of the head.
- If you are using setup B:
  - o Attach the magnet to the edge of the holder as shown in Figure 4B.
  - o Place the magnet approximately 5 cm from the tip of the NV magnetometer, ensuring it is aligned with the central axis of the head.
- Click “Run Measurement”
- If the two ODMR dips are not clearly visible, adjust the magnet position (typically moving it further away).
- Once both dips are clearly visible within the frequency sweep range, click “Save Data”. You can find your spectrum (an Excel file) in the Downloads folder.
- Increase the magnet distance by approximately 1 cm and repeat the measurement. Collect a total of 6 to 8 measurements, each at a different magnet distance.

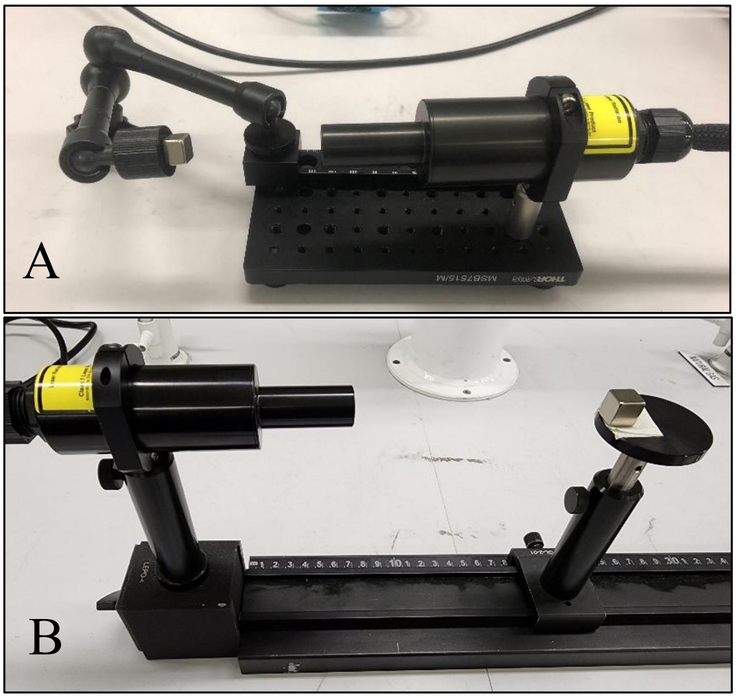


Figure 4: Photographs of two magnet setups (A and B), showing the magnet positioned along the central axis of the NV magnetometer head to induce Zeeman splitting.

Part 2. Calculating magnetic field

- For each measurement, plot the recorded ODMR spectrum and identify the centre frequency of each dip.

- Calculate the Zeeman splitting by subtracting the two dip frequencies and dividing the result by 2.
- Use Equation (5) to convert the Zeeman splitting into magnetic field strength. Record the calculated field strength corresponding to each distance from the magnet.
- Include representative ODMR spectra (with clearly marked dips) in your portfolio.
- Plot magnetic field strength (B) versus distance (d) from the magnet and fit the data to the power-law relationship:

$$B \propto d^{\alpha}$$

- Extract the exponent α from the fit. Comment on how your result compares with theoretical expectations (e.g., $\alpha \approx -3$ for a magnetic dipole).
- Briefly discuss the potential applications of spin-based quantum sensing technologies, such as nanoscale magnetometry and biomedical imaging.

BONUS: For improved precision, use software such as OriginLab or Python to fit each ODMR spectrum using two Lorentzian functions and extract the center frequencies from the fitted curves.

References

1) M. Ruf, H. Choi, D. Englund, R. Hanson, *J. Appl. Phys.* 130, 070901 (2021)
2) A. Kuwahata, T. Kitaizumi, K. Saichi, T. Sato, R. Igarashi, T. Ohshima, Y. Masuyama, T. Iwasaki, M. Hatano, F. Jelezko, M. Kusakabe, T. Yatsui, M. Sekino, *Sci. Rep.*, 10, 2483 (2020)

Experiment 1B: The Hall Effect and Semiconductor Properties

Aims

In this experiment, you will use the Hall effect to characterise carriers in germanium (Ge). In particular, you will study the transport of majority carriers in the presence of a magnetic field, determine the carrier type and concentration, carrier mobility and the Fermi energy.

Equipment list

- Hall effect apparatus with a germanium sample
- Magnets
- Current supply
- Resistance box
- Digital multimetres
- Gauss meter

Introduction

The Hall effect relies on the presence of a magnetic field through a current carrying conductor. In this case, the force due to the magnetic field on a moving charge is,

$$\boldsymbol{F}_M = q\boldsymbol{v} \times \boldsymbol{B}, \tag{1}$$

where q is the carrier charge, $\boldsymbol{v}$ is the drift velocity of the carrier and $\boldsymbol{B}$ is the magnetic field. In semiconductors, the charge carriers are either holes (p-type) or electrons (n-type). For a magnetic field directed into the page, we have the situation shown in Fig. 1.

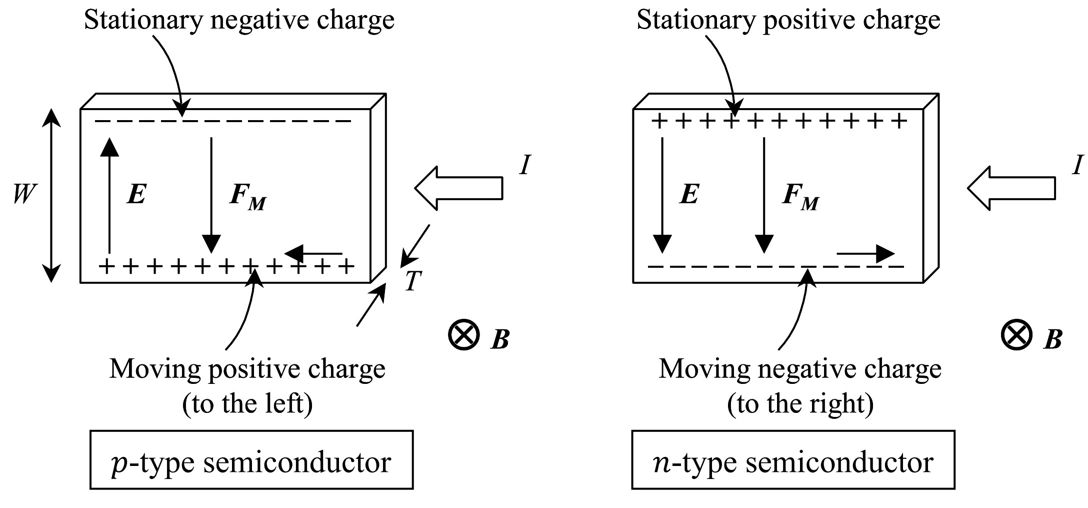


Fig. 1. The behaviour of n-type and p-type semiconductors in a magnetic field. Note that the magnetic forces, F_M, in two cases are in the same direction, but the direction of Hall field (E) is the opposite.

The charge will separate as shown in Fig. 1, setting up an electric field that will oppose the separation. There will come a point in the separation where the force due to the electric field, E, arising from the separation and that giving rise to the separation, namely $\boldsymbol{F}_M$, become equal in magnitude and opposite in direction, so that,

$$\boldsymbol{F}_M = q\boldsymbol{E},$$
$$q\boldsymbol{vB} = q\boldsymbol{E},$$
$$v = \frac{E}{B}. \tag{2}$$

The longitudinal current density is given by $J = nqv$, where n is the density of mobile charge carriers in the material (the quantity we are to find here by experiment) and q their charge. If the sample has thickness T and width $W,$ then the cross-sectional area of the conductor is $A = WT$ and the current $I = JA$.

The Hall constant, R_H, is defined as

$$R_H = \frac{E}{BJ}. \tag{3}$$

Combining equations (2) and (3), and noting that $J = nqv$, we get

$$R_H = \frac{v}{J} = \frac{1}{nq}. \tag{4}$$

Hence the Hall constant is the drift velocity of the carrier per unit current density. Furthermore, from the definition of the carrier mobility μ, $v = \mu E_L$, and the longitudinal conductivity $\sigma, J = \sigma E_L$, where E_L is the longitudinal electric field,

$$R_H = \frac{\mu}{\sigma}. \tag{5}$$

Rearranging equation (4) gives

$$n = \frac{1}{qR_H} = \frac{BJ}{qE} = \frac{BI}{qEWT}. \tag{6}$$

Now the electric field E is related to the Hall voltage, V_H, by $E = V_H/W$, so that,

$$n = \frac{BI}{qTV_H}. \tag{7}$$

Equation (7) allows the calculation of the charge carrier density of a conductor by measurement of the Hall voltage as a function of current.

Experimental Work

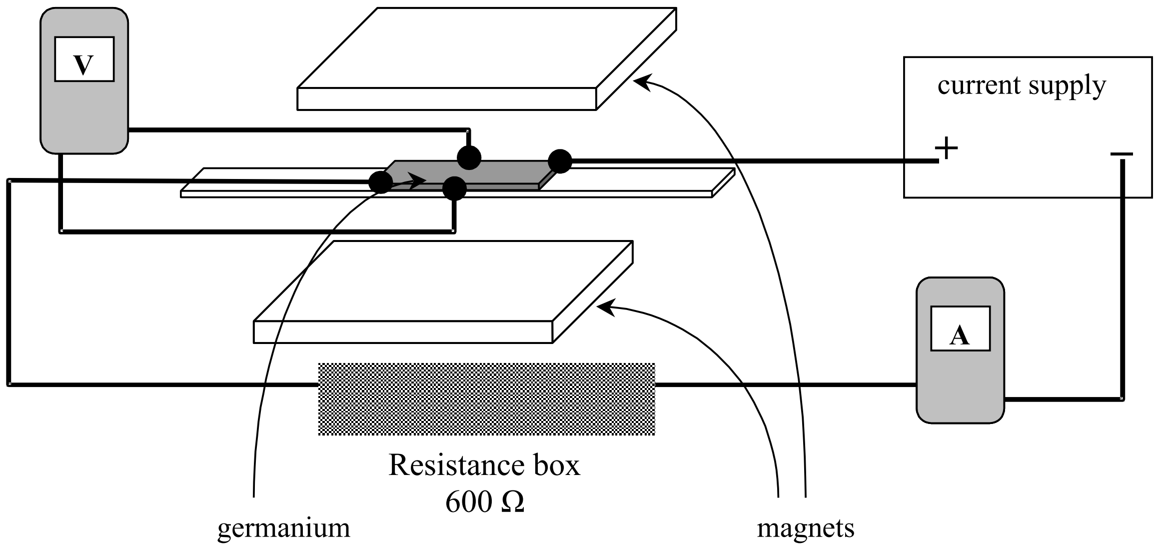


Fig. 2. Experimental apparatus for measurement of the Hall voltage.

(a) First, you need to record (or measure if not available) the length and width of the germanium (Ge) sample using a ruler. The thickness of the sample is 1 mm.

Caution

This experiment uses very strong magnets. They should be handled with care and it is recommended that you wear safety glasses.

(b) Determination of the majority carrier type

Caution

Always use the resistance box (set to 600 Ω) to limit the current through the sample and never exceed 50 mA, otherwise the sample will be damaged.

The equipment setup is shown in Fig. 2. It is suggested that you draw in your portfolio a schematic diagram of your experimental setup with sample polarities and the direction of current passing through the sample.

Sketch a block diagram to show the magnetic field, the direction of current flow and Lorentz force for your setup. Use this diagram to determine the type of majority carriers (i.e. positively or negatively charged) in your Ge sample.

(c) Measurement of the Hall voltage as a function of current in the magnetic field.

Use the potentiometer on the circuit board to zero out the voltage across the sample at zero magnetic field, i.e. with the magnets absent. Make sure this voltage is zero even when a high current is passing through the sample.

Question: Why does this voltage arise even at zero magnetic field?

Without changing the potentiometer setting, place the Ge sample and the magnets so that the sample is at the centre of the magnet gap. Take measurements at various longitudinal currents, but never exceed 50 mA. You need to record the applied voltage, current and Hall voltage.

(d) Reverse the polarity of the current supply with respect to the connections of the sample and repeat the measurements as described in (c). Again, you need to reset the potentiometer at zero field before taking measurements.

(e) Measure the strength of the magnetic field in the gap between the magnets using a Gauss meter. For accurate measurements, you need to keep the separation between the magnets the same as when they are attached to the Ge sample. *For accurate magnetic field measurements, the magnetic flux must enter the Hall element on the Gauss meter probe at the right angle.*

Caution

The Gauss meter probe is very fragile, please use with special care.

Analysis

(a) Plot the Hall voltage as a function of the current through the germanium and calculate the density of majority carriers.

(b) Having obtained the majority carrier density, determine the minority carrier concentration and the position of the Fermi level E_F in the band gap. Draw the energy band diagram for the Ge sample including the position of its Fermi level relative to the band edges.

Hint: You will need to look up fundamental parameters for Ge semiconductor.

(c) Determine the Hall constant R_H (see equations (3)-(7)). Plot appropriate graphs to determine the conductivity of the Ge sample and calculate the mobility of the majority carriers.

(d) Comment on your results and discuss sources of uncertainty.

Experiment 2: Photoconductivity

Aim

To investigate the photoconductive properties of CdS. These properties are governed by several factors including the intensity of incident light, excitation wavelength, photoconductive gain, carrier lifetime and trapping.

Equipment List

- Laser diode
- Detector
- Laser power meter
- Polariser
- Adjustable iris
- CdS photoconductive cell
- Black cloth or optical isolator
- Rigol digital storage oscilloscope (DSO)
- Microscope camera

Theoretical Background

Basic concepts

Some semiconductors, for example, CdS, CdSe, Ge, Si, PbS, Se and the inter-metallic AB compounds are insulators in the dark with only a few carriers in the conduction band (i.e., thermally excited across the gap). However, they become conductors when illuminated by light because absorbed photons create free holes and electrons (see Fig. 1).

The conductivity is given by:

$$\sigma = e\left(n\mu_n + p\mu_p\right)\,\Omega^{-1}m^{-1}, \tag{1}$$

here e is the electronic charge, n and p are the densities of free electrons and holes, and μ_n and μ_p are their mobilities respectively. In most photoconductors, the two mobilities differ by several orders so that the conduction occurs by one type of carrier. If n-type carriers dominate then Eq. (1) reduces to

$$\sigma = ne\mu_n, \tag{2}$$

With $\mu_n = v_D/E$ where v_D is the electron drift velocity and E the applied electric field. The resistance of a piece of photoconductive material of length d and cross-sectional area A is

$$R = \frac{1}{\sigma}\frac{d}{A}. \tag{3}$$

If the applied voltage is V, then the photocurrent is given by

$$i = \frac{V}{R} = \frac{VA}{d}\sigma. \tag{4}$$

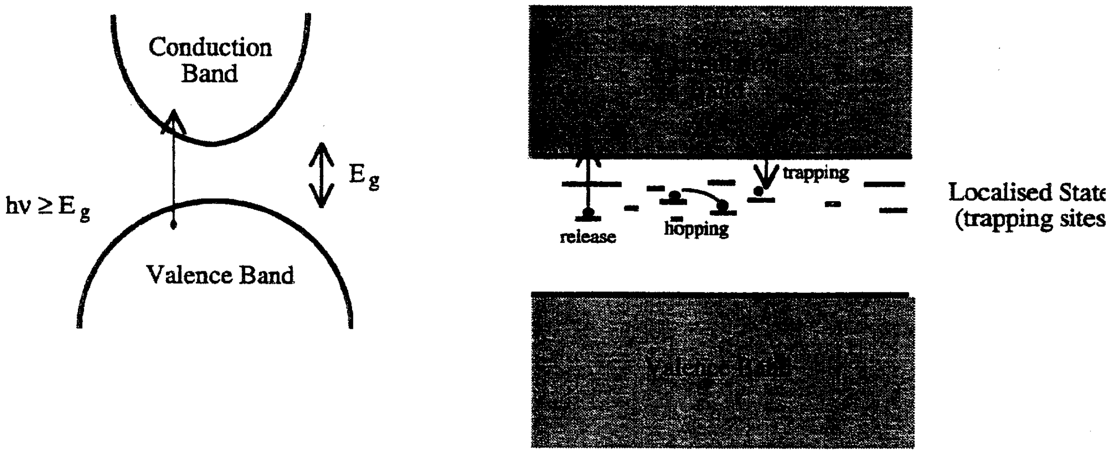
The change in conductivity in these materials when illuminated makes them useful as detectors of infrared radiation, as optoelectronic switches and in photocopying processes.

Role of the bandgap in a semiconductor

Excitation across the bandgap of a semiconductor occurs only by a laser whose photon energy hv exceeds the energy gap E_g. This energy determines the longest wavelength for bandgap absorption.

$$hv > E_g \tag{5a}$$

$$\text{or} \quad \lambda < \frac{hc}{E_g} \tag{5b}$$



(a) The production of carriers

(b) Schematic representation of trapping and release of carriers at defects

Fig. 1. Process of photoconductivity in semiconductors.

The role of the energy gap and a representation of the processes occurring during the absorption of a photon are illustrated in Fig. 1. The long wavelength limit (λ_{max}) of some photoconductive materials are given in Table 1. However, defects can also create states in the bandgap and these states can potentially absorb sub-bandgap photons and affect the photoconductivity properties of a semiconductor.

Table 1. Wavelength limit for different photoconductive material

Material	λ_{max} (nm)	Type
CdS	510	intrinsic
Se	620	intrinsic
GaAs	920	intrinsic
Si	1100	intrinsic
Ga	1800	intrinsic
Pb	3100	intrinsic
InSb	7700	intrinsic
GeCu	29000	extrinsic
GeZn	38000	extrinsic

The number of electrons contributing to current per absorbed photon is called G, the photoconductive gain. Each absorbed photon produces one electron; thus, the photoconductive gain is given by the number of carriers passing a point per second (i/e) divided by the number of photons absorbed per second (FdA), i.e.

$$G = \frac{i}{e}\frac{1}{FdA}. \tag{6}$$

Here, F is the number of photons absorbed per unit volume and $d.A$ the volume of photoconductor material.

Thus, the photocurrent is

$$i = eFdAG. \tag{7}$$

F depends on the light intensity. Assuming all incident light is absorbed, we can write F as

$$F = \frac{I}{h\upsilon t}, \tag{8}$$

where t is the thickness of the photoconductor and I intensity of incident light (in W/m^2).

Referring to Fig. 4 to measure the area of the semiconductor and substituting (8) into (7) we obtain

$$i = \frac{edlG}{h\upsilon}I. \tag{9}$$

The photocurrent i *does not, in practice, increase linearly with the optical intensity* I *because* G *itself depends on the intensity of incident light* I. Thus,

$$i = KI^{\alpha}, \tag{10}$$

and α is usually between 0.5 and 1. Both super $(\alpha > 1)$ and sub-linear $(\alpha < 1)$ behaviours are possible.

Carrier lifetime and photoconductive gain

Whether a photo-produced carrier contributes to current flow or not depends on whether it can traverse the distance between electrodes before being lost. Loss of carriers in semiconductors occurs via recombination. In photoconductors the carrier density, n, depends on F, the number of carriers produced by the light, and carrier lifetime τ, so we can write

$$F = \frac{n}{\tau}. \tag{11}$$

The *carrier lifetime* τ is not to be confused with trapping times. It is the lifetime of the electron excited across the energy gap into the conduction band.

Recombination often occurs with trapped carriers or carriers which are held at some trapping site, typically a lattice defect. However, trapped carriers can also be thermally re-emitted from a trapping site to the conduction or valence band, so they are not necessarily lost as carriers. Also, the drift velocity is

$$v_D = \mu E = \frac{\mu V}{d}, \tag{12}$$

and the transit time of carriers between electrodes is

$$t_{tr} = \frac{d}{v_D} = \frac{d^2}{\mu V}. \tag{13}$$

Substituting equations (4) and (11) into (6), we get

$$G = \frac{\tau \mu V}{d^2} = \frac{\tau}{t_{tr}}. \tag{14}$$

Thus, the photoconductive gain is simply the ratio of the carrier lifetime to the transit time.

Experimental Work

Laser Safety Caution

Never look directly into a laser beam as it can cause severe eye damage.

Part 1. Calibration of laser intensity

The apparatus for this section is set up as in Fig. 2(a). The red laser diode ($\lambda = 633$ nm) must be calibrated before any meaningful measurements can be done. With this laser the CdS cell is excited with a photon energy $h\nu = 1.96$ eV, which is below the CdS bandgap of 2.43 eV.

Mount the detector head of the power meter at the location where the photoconductive cell is to be placed. The experimental arrangement for the laser calibration is shown in Fig. 2(a). The light intensity can be varied using the polariser.

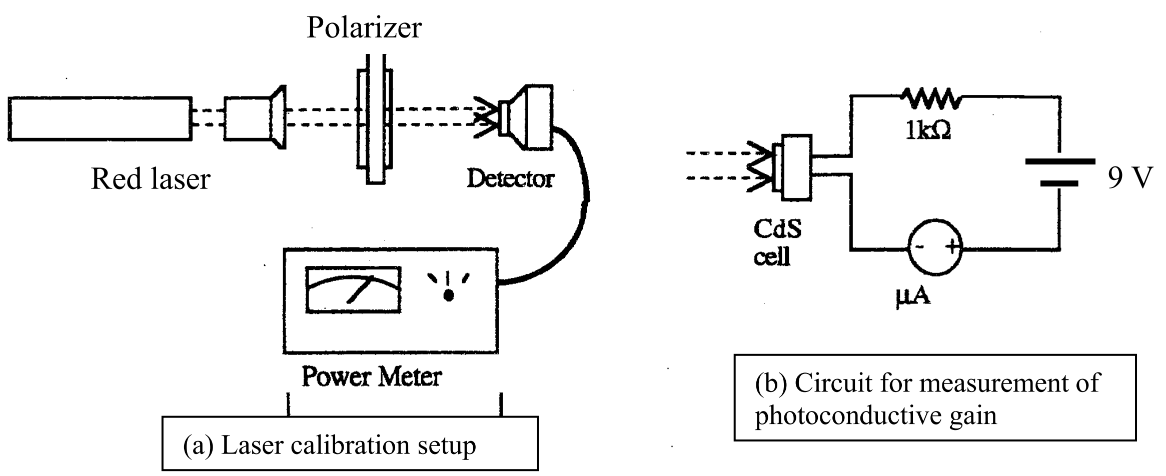


Fig. 2. Experimental apparatus to measure photoconductivity.

If P is the power detected and A is the beam area, then the light intensity I is (assuming the intensity is uniform over the area):

$$I = \frac{P}{A}. \tag{15}$$

Measure the beam area at the detector location.

Take readings of power P versus polarizer angle θ in the range $\theta = 0 - 180°$. You need to cover the apparatus to block out any stray room light using a black cloth or optical isolator while taking measurements. Plot I versus θ as a calibration curve.

Part 2. Photoconductive properties of CdS
In this part, you will study elementary photoconductive properties of the CdS cell. All the photoconductive measurements are carried out using the circuit of Fig. 2(b) with the CdS cell replacing the detector in the laser beam.

A. Measure the dimensions of the CdS cell using a microscope camera (see Appendix)

Values required are:

- Interelectrode distance d, i.e. the width of the CdS photoconductor.
- The total length of the photoconductor exposed to light, l.
- The thickness of the CdS photoconductor is $t = 0.12$ mm.

From the measurements, you can calculate the cross-section area of the photoconductor (perpendicular to the conduction path): $A = lt$

B. Measure the photoconductive gain and the conductivity
With the CdS cell in position in the laser beam and using the circuit of Fig. 2(b), record the photocurrent i at *a minimum of 8 different light intensities over the excitation power range* measured in Part 1.

Using an appropriate log-log plot, find α of equation (10). Comment on the significance of your results.

Using equation (9) find the photoconductive gain G and plot G as a function of light intensity. Discuss the significance of your results.

Part 3. Measurement of minority carrier lifetime
The setup for this section is shown in Fig. 3. *Using the data you have collected in Part 1, set the polarizer angle so that the laser intensity is close to maximum.*

As the mobility of electrons in CdS is considerably greater than that of holes, the photocurrent is due mainly to the motion of electrons. However, the decay of the photocurrent is controlled by minority carriers i.e. holes that are trapped in defect states. When the photoexcitation is turned off, trapped holes are thermally released, electron-hole recombination occurs and the photocurrent decays. The photocurrent at a time, *t*, after the extinction of the illumination can be written as:

$$i = i_{\max} e^{-\frac{t}{\tau_m}} \tag{16}$$

where τ_m is the minority carrier lifetime.

To measure τ_m you need to obtain a photocurrent decay curve on the oscilloscope. The lifetime τ_m is equal to the time over which the photocurrent falls to 1/*e* (or 37%) of its initial value. You can estimate τ_m from the displayed decay curve. Repeat the lifetime measurement until successive measurements agree within 10%. Alternatively, you can export data from on the storage oscilloscope; equation (16) can be linearised and you can plot an appropriate graph to determine τ_m from the decay side of the photocurrent curve.

Discuss the significance of your lifetime measurement results.

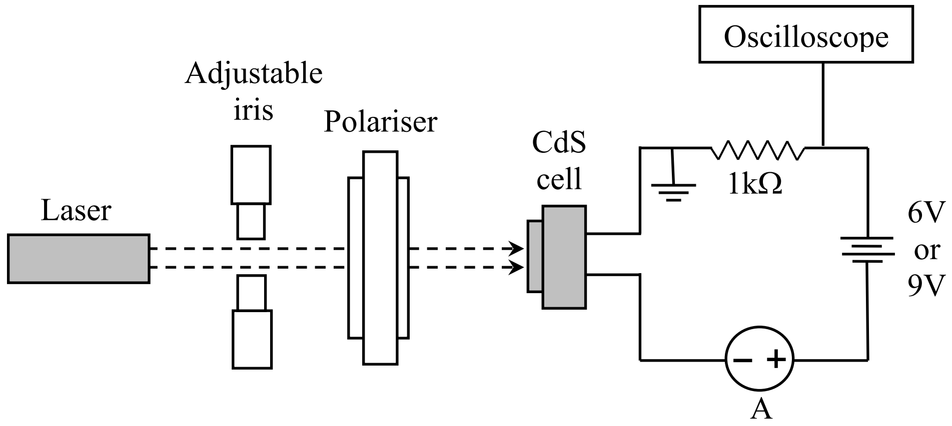


Fig. 3. Setup for measuring carrier lifetime.

Operating instructions for capturing a photocurrent decay curve
To measure τ_m you need to obtain photocurrent decay curves. Set the horizontal time scale on the DSO approximately between 1 to 10 ms. Move the trigger level to ~ 70% of the DC signal using the "Trigger" dial. Select "down" for the slope option and make the sweep "single". Setting a single sweep will stop the DSO from recording new data so that the single-shot event is captured by the DSO. Once configured, block the laser beam with a piece of paper and the decay curve should be displayed on the DSO screen.

Part 4. Above-bandgap excitation of CdS
Replace the red laser with the violet laser ($\lambda = 406$ nm) and repeat the measurements as described in parts 1-3. Under the excitation with the violet laser, the CdS cell is excited with a photon energy $\boldsymbol{h\nu}$ = 3.05 eV, which is above the CdS bandgap.
Discuss the photocurrent response of the CdS cell under the red and violet laser excitation regimes and the influence of excitation wavelength on the photoconductivity properties of CdS.

APPENDIX

Measure the dimensions of the photoconductor in the CdS cell using a microscope or digital camera. You need to measure:

- Inter-electrode distance, d
- The total length of the photoconductor, l
- The thickness of the CdS ($t = 0.12$ mm)

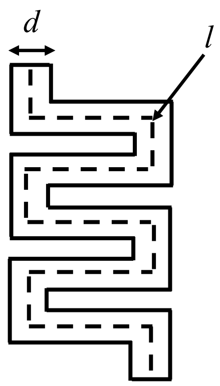


Fig. 4. Simplified diagram of the CdS photoconductor.

Experiment 3: Measurement of the band gap in semiconductors

Aims

- To determine the band gap in several semiconducting materials using the temperature dependence of the electrical conductivity of p-n junctions.
- Conduct regression analysis using scientific analysis software such as OriginLab or any advanced package (which is not Excel).
- Identify the material in p-n junctions.

Equipment List

- Digital voltmeters
- Yokogawa DC source
- 5 constant current sources
- Diode board (contains different p-n junction diodes and Pt resistance thermometer)
- Liquid nitrogen
- Protective gloves and safety goggles

Theoretical Background

The band gap in semiconductors is the central feature in the explanation of the electrical behaviour of solid-state devices. The band gap of a semiconductor is temperature dependent, and the values of representative semiconductors at 0 K and 300 K are given in Table 1. In this experiment, you will study the band gap in a semiconducting material using the temperature dependence of the electrical conductivity of a p-n junction which is made of the semiconductor.

Table 1. Band gaps of representative semiconductors.

Semiconductor	Energy gap type	E_g (T=0 K) (eV)	E_g (T=300 K) (eV)
Si	indirect	1.17	1.11
Ge	indirect	0.74	0.67
AlAs	indirect	2.24	2.16
GaP	indirect	2.32	2.26
GaAs	direct	1.52	1.43
GaSb	direct	0.81	0.78
InP	direct	1.42	1.35
InAs	direct	0.43	0.36
GaN	direct	3.45	3.40

The ideal current-voltage characteristic of a semiconductor p-n junction is given by the so-called diode equation

$$I = I_o \left[\exp\left(\frac{eV}{kT}\right) - 1\right], \tag{1}$$

where I_o is the saturation current when the junction is reverse biased, e is the electronic charge, k the Boltzmann's constant, T the temperature of the junction in Kelvin, and V the potential difference across the junction. I_o was shown to be proportional to the square of the intrinsic carrier concentration, n_i^2. If the doping profile across the junction shows a sharp discontinuity, the saturation current I_o can be written as

$$I_o = AT^{(3+\gamma/2)} \exp\left[-\frac{E_g(T)}{kT}\right], \tag{2}$$

where A is a constant, $E_g(T)$ the band gap between the valence and conduction band, γ a constant depending on the temperature dependent transport properties of the minority carriers.

As the temperature increases, the band gap decreases. The relationship between the band gap and temperature, to the first approximation, can be written

$$E_g(T) = E_g(0) - aT, \tag{3}$$

where $E_g(0)$ is the limiting value of the band at 0 K, a is a constant. $E_g(T)$ is in units of electron volts. If it is assumed that $eV \gg kT$, combining equations (1), (2) and (3), taking natural logs and rearranging we get

$$eV = E_g(0) + \left[k \ln\left(\frac{I}{A}\right) - a\right]T - k\left(3 + \frac{\gamma}{2}\right)T \ln T. \tag{4}$$

The above equation can be rewritten as

$$eV = E_g(0) + \alpha T - \beta T \ln T, \tag{5}$$

where $\alpha = k \ln\left(\frac{I}{A}\right)$ and $\beta = k\left(3 + \frac{\gamma}{2}\right)$.

Throughout the duration of the experiment the current I, and hence α, is kept constant, and the potential difference across the junction, V, is measured as a function of temperature. Since Eq. (5) is linear in $E_g(0)$, α, and β, it is possible to determine their optimal values from a linear least-squares fit to the experimental data.

Experimental Work

Caution

This experiment uses liquid nitrogen. Contact with skin can result in severe frostbite. Always wear cryogenic gloves and safety goggles when handling liquid nitrogen.

(a) Connect the Pt resistance thermometer to the Yokogawa DC source. Set the output current to 1.0 mA. Connect a digital voltmeter in parallel with the thermometer.

The thermometer has been previously calibrated and was found to fit the following function:

$$T = 31.32 + 2.211R + 2.899 \times 10^{-3}R^2 - 1.004 \times 10^{-5}R^3 + 1.687 \times 10^{-8}R^4. \quad (6)$$

Use this equation to calculate the temperature and verify that the result is reasonable for room temperature conditions.

(b) Measure the current output of each constant current source to confirm it matches the specified value. Set the input voltage to approximately 15 V and verify that the output current is 0.12 mA.

(c) Connect each of the diodes in the diode board to a constant current source. A voltmeter should be connected in parallel with each diode. The voltage across the junction at room temperature should be between 0.15 and 0.6 V for the Si and Ge junction, and between 0.9 and 2 V for the remaining diodes.

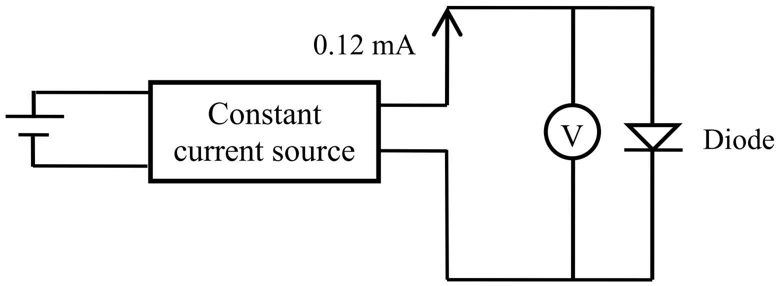


Fig. 1. Schematic diagram of a diode connected to a constant current source of 0.12 mA.

(d) Carefully pour liquid nitrogen into the metal container. (*Caution:* Do not pour liquid nitrogen directly onto the diode board, as rapid thermal contraction may cause cracking or damage.)

Record the *voltage across all the diodes as well as the Pt resistance thermometer* as the board gradually warms up to room temperature.

Continue taking measurements at regular intervals, ensuring a sufficient number of data points across the full temperature range. You may also consider recording a video or taking photos of the experiment using your phone.

(e) Repeat the procedure described above for a second set of measurements. To simplify averaging, aim to record the second set of voltage values at approximately the same temperature points as in the first run.

Analysis

(a) Using your recorded values of voltage and temperature, plot eV versus temperature for each of the p–n junctions. The measured voltage V across each junction corresponds to the electron energy eV, where:

$$eV\text{ (in eV)} = V\text{ (in volts)}$$

This is because 1 volt corresponds to 1 electronvolt of energy when the charge is that of a single electron (i.e., $e = 1.6 \times 10^{-19}$ C). In other words, if the voltage across the junction is 0.65 V, then the corresponding electron energy is 0.65 eV.

(b) The temperature-dependent energy is modelled by the empirical expression:

$$eV = E_g(0) + \alpha T - \beta T \ln T$$

Perform a non-linear regression analysis to fit your data to the equation above. Extract the best-fit values and uncertainties for the parameters:

- $E_g(0)$: band gap at 0 K
- α: temperature coefficient
- β: logarithmic temperature coefficient

Refer to the Appendix for general guidelines on fitting and analysis.

(c) Generate plots showing both the measured data points and the fitted curves for each junction. Adjust the temperature range if needed to improve the quality of the fit. Inspect the fit for deviations and comment on the validity of the model across the measured temperature range.

(d) Report the extracted band gap energy $E_g(0)$ along with its uncertainty for each p–n junction. Compare your measured band gaps to known values for common semiconductors (e.g., Si, Ge, GaAs) and attempt to identify each junction type.

(e) Discuss possible sources of uncertainty, including instrument accuracy and fitting errors.

Appendix: Regression Analysis

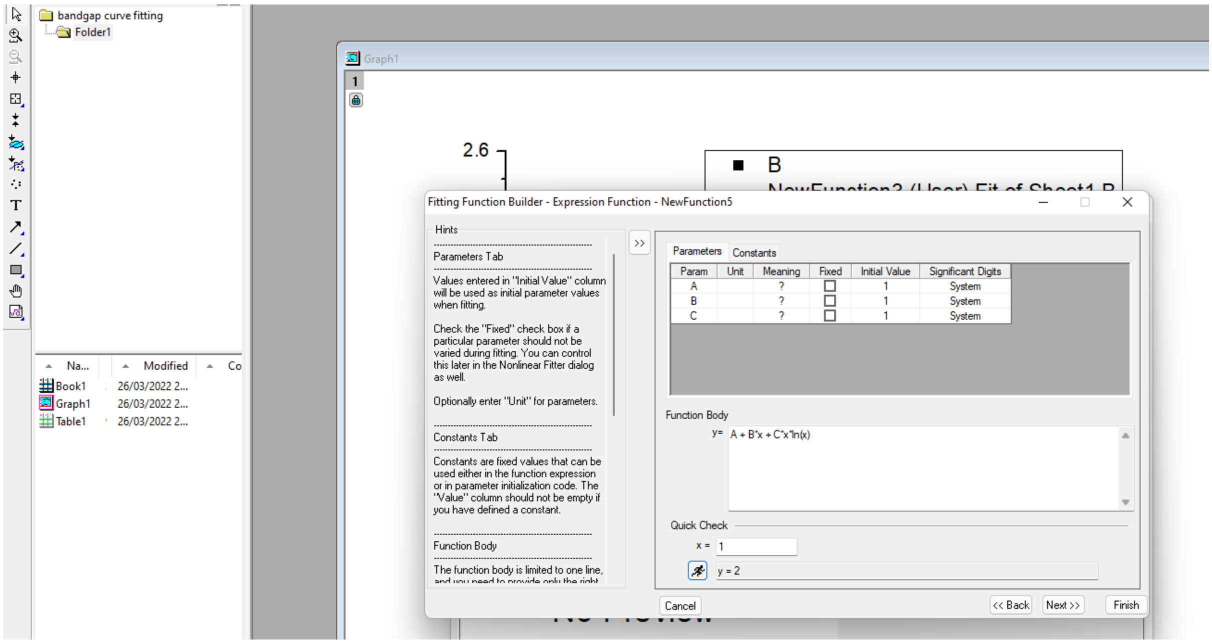
You should use scientific software such as OriginLab to perform regression analysis. Below are general guidelines.

See instructions on how to fit a user-defined fitting function in the link below.
https://www.originlab.com/doc/tutorials/userdef-fitfunc

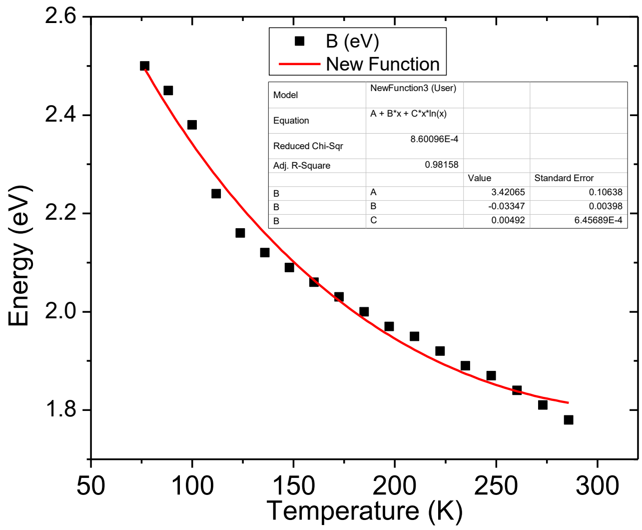
Eq. (5) is not a linear function of T and can be expressed as:

$$y = A + Bx + Cxln(x) \qquad (7)$$

In Origin version 8, create a workbook that contains the variable y (representing eV) versus x (representing T), then highlight these columns and generate a scatter plot. In the next step, select Analysis → Fitting → Nonlinear curve fit → Open Dialog box. In this box, enter the fitting equation in the **y =** box with three fitting parameters A, B and C. Here the parameter table will display the initial values for the fitting parameters.



Once the equation has been entered, click Fit button to fit the data set to the function you have just created. The fitted curve will be displayed as a continuous line in the plot area.



Note: You can download the student version in the below link. This version is fully functional and lasts for six months. Make sure you finish the data analysis before your license expires (unless you want to purchase it).
https://www.originlab.com/index.aspx?go=Products/OriginStudentVersion